

MOTIONS		
w	: b	Next / prev word
W	: B	Next / prev WORD
e	: ge	End of word / prev end
f{c}	: F{c}	Jump to char fwd / bwd
t{c}	: T{c}	Before char fwd / bwd
;	: Ctrl+\	Repeat f/t fwd / bwd
O	: ^ : \$	Line start / first char / end
gg	: G : {n}G	File start / end / line n
{	: }	Prev / next blank line
[[	:]]	Prev / next section
H	: M : L	Screen top / mid / bottom
j	: k	Display line down / up
Ctrl+D	: Ctrl+U	Half-page down / up
Ctrl+F	: Ctrl+B	Full page down / up
Ctrl+E	: Ctrl+Y	Scroll one line (cursor stays)
zz	: zt : zb	Center / top / bottom cursor line

JUMPING		
gf		Go to file under cursor
gd		Go to local def (LSP overrides)
Ctrl+]		Jump to tag (tags)
Ctrl+T		Jump back from tag
Ctrl+O	: Ctrl+I	Jumplist back / fwd
g:	: g,	Changelist older / newer
gi		Last insert pos & enter insert
%		Matching bracket (matchup)
g%		Match in reverse direction
[X	:]%	Prev / next outer block
z%		Jump inside nearest block

SEARCH & REPLACE		
/	: ?	Search fwd / bwd
n	: N	Next / prev match (with pulse)
*	: #	Word under cursor fwd / bwd
gn	: gN	Select next / prev search match
Space		Clear search highlighting
:%s/old/new/g		Replace all in file
:%s/old/new/gc		Replace all (confirm each)

FZF FUZZY FINDER		
Ctrl+P	: ,f	Find files (with preview)
,b		Buffers
,l		Lines in open buffers
,G		Ripgrep search (live)

INSIDE FZF POPUP

Ctrl+T	: X : V	Tab / hsplit / vsplit (popup only)
Ctrl+/,		Toggle preview window

UNMAPPED FZF COMMANDS

:History		Recent files
:Tags		All tags
:Commands		All Vim commands

EDITING

ENTERING TEXT

i	: a	Insert before / after cursor
I	: A	Line start / end (v-block: each line)
o	: O	New line below / above
r	: R	Replace char / replace mode

VISUAL SELECTION

v	: V : Ctrl+V	Char / line / block visual
gv		Reselect last selection
o		Other end of selection

OPERATORS*

c{motion}*	: C : cc	Change motion / EOL / line
d{motion}*	: D : dd	Delete motion / EOL / line
y{motion}*	: yy	Yank motion / line
y	: Y (visual)	Yank selection / to clipboard
p	: P	Paste after / before

UNDO, REPEAT & JOIN

u	: Ctrl+R	Undo / redo
.		Repeat last change
@:		Repeat last Ex command
&		Repeat last :substitute
J	: gJ	Join lines / without space

CASE, INDENT & FORMAT

-		Toggle case (char)
gu{motion}*	: ! gu{motion}*	Upper / lowercase motion
u	: U (visual)	Lower / uppercase selection
>>	: <<	Indent / dedent line
<	: > (visual)	Indent / dedent & keep selection
=	: {motion}* : ==	Auto-indent motion / line
gq{motion}*		Format text over motion
!{motion}*	: {cmd}	Filter through shell cmd
Ctrl+A	: Ctrl+X	Increment / decrement number
,=		Reformat entire file

COMMENTS & CUSTOM

gcc	: gc{motion}*	Comment line / motion (tcomment)
,/	: ,ci	Toggle comment (normal + visual)
,sa		Select all
,U	: ,u	Upper / lowercase word

INSERT MODE

Ctrl+O		Execute one normal cmd, return
Ctrl+R {reg}		Paste register content
Ctrl+W		Delete word bwd
Ctrl+U		Delete to line start
Ctrl+T	: Ctrl+D	Indent / dedent current line
Ctrl+N	: Ctrl+P	Completion next / prev
Ctrl+X	: Ctrl+F	Filename completion
Ctrl+X	: Ctrl+L	Whole line completion
Ctrl+X	: Ctrl+O	Omni completion (LSP)

TEXT OBJECTS

Use with operators* — i=inner, a=around

BUILT-IN

iv	: av	Word
is	: as	Sentence
ip	: ap	Paragraph
i"	: i' : i`	Quoted string
i{	: i{ : i[: i<	Bracket pair
it	: at	HTML/XML tag

TARGETS.VIM

i,	: a,	Comma-separated item
i;	: i= : i+	Any separator item
ia	: aa	Function argument
ib	: ab	Any block () {} []
iq	: aq	Any quote ' " `
in{	: il{	Next / last pair
ina	: ila	Next / last argument

CUSTOM

i_	: a_	Underscore segment
if	: af	Function
iz	: az	Fold
i%	: a%	Matching block (matchup)

SURROUND

ys{motion}{c}		Add: ysiw" → "word"
yss{c}		Surround entire line
cs{old}{new}		Change: cs" → "word"
ds{c}		Delete: ds" → word
S{c}		Surround selection (visual)
,"	: ',	Quick-quote word

MARKS, REGISTERS & MACROS

m{a-z}		Set mark
'{a-z}	: `{a-z}	Jump to mark line / exact pos
''	: ``	Jump to last position
"{r}y	: "{r}p	Yank / paste register r
"+	: "+ : "O	Clipboard / selection / last yank
q{a-z}	: q	Record / stop macro
@{a-z}	: @@	Play macro / replay last

FOLDING

za		Toggle fold
zo	: zc	Open / close fold
zR	: zM	Open all / close all folds
zj	: zk	Next / prev fold

SHELL & MISC

!: {cmd}		Run shell command
:r !{cmd}		Insert cmd output into buffer
:w !{cmd}		Pipe buffer to command
q:	: q/	Cmd / search history window
Ctrl+R	: Ctrl+W	Word at cursor (cmd line)
ZZ	: ZQ	Save+quit / quit without save
Ctrl+Z		Suspend Vim (fg to return)
Ctrl+G	: Ctrl+L	File info / redraw screen

* OPERATOR + MOTION / TEXT OBJECT COMBINATIONS

Operators define **what** to do, motions/text objects define **where**. Any operator (`c d y > < = gU gu gq gc !`) combines with any motion or text object from the sections above. Doubling applies to the whole line: `dd yy cc >> << == gcc`.

DELETE `dv` (word), `d$` (EOL), `dd` (line), `di"` (in quotes), `dip` (paragraph), `d/foo` (until match)

CHANGE `cw` (word), `c1w` (inner word), `ci"` (in quotes), `ci{` (in parens), `cc` (line), `ct,` (until ,)

YANK `yv` (word), `y$` (EOL), `yy` (line), `y1{` (in braces), `yip` (paragraph), `yG` (to EOF)

INDENT `>>` (line), `>ip` (paragraph), `>i{` (in braces), `=` (auto-indent), `=ip` (paragraph)

CASE `gUiv` (upper word), `guiw` (lower word), `gU$` (upper to EOL)

COMMENT `gcc` (line), `gcip` (paragraph), `gcif` (in braces), `gc3j` (3 lines)

FORMAT `gqip` (format paragraph), `gqi{` (in braces), `!ip sort` (sort paragraph)

OTHER `dG` (to EOF), `ygg` (to start), `d3j` (3 lines), `d%` (to match)

VIM CHEAT SHEET — PLUGINS, LANGUAGES & MORE

Leader , Local Leader - (used by vimtex)

LSP (C/C++, PYTHON, FORTRAN)		
<code>gd</code> <code>gD</code>	Go to / peek definition	
<code>gr</code>	Find references	
<code>K</code>	Hover documentation	
<code>rn</code>	Rename symbol	
<code>,ca</code>	Code action	
<code>[d</code> <code>]d</code>	Prev / next diagnostic	

CODE NAVIGATION		
<code>,j</code>	Jump to def/usage (AnyJump)	
<code>,i</code>	Header ↔ source (C/C++)	
<code>,ff</code>	Clang-format buffer (C/C++)	
<code>,.</code>	Alternate file (test ↔ source)	
<code>:A</code> <code>:AS</code> <code>:AV</code>	Alternate / split / vsplit	
<code>:AT</code>	Alternate in new tab	
<code>:HeaderguardAdd</code>	Add header guard (C/C++)	
<code>:ClangFormatAutoToggle</code>	Toggle auto clang-format	

BUILD & DISPATCH		
<code>,mm</code>	Async :Make (background)	
<code>,md</code>	:Dispatch command	
<code>:Start {cmd}</code>	Run interactive command	
<code>:Spawn {cmd}</code>	Run in background	
<code>:Focus {cmd}</code>	Set default dispatch target	
<code>:AbortDispatch</code>	Abort running dispatch	

DEBUGGING (VIMSPECTOR)		
<code>,dd</code> <code>,de</code>	Launch / reset debugger	
<code>,db</code> <code>,dB</code>	Breakpoint / conditional	
<code>F5</code>	Continue / pause	
<code>F9</code>	Toggle breakpoint	
<code>F10</code>	Step over	
<code>F11</code> <code>F12</code>	Step into / step out	

SNIPPETS & COMPLETION		
<code>Ctrl+J</code> <code>Ctrl+K</code>	Expand snippet & jump fwd / bwd	
<code>Tab</code> <code>Shift+Tab</code>	Cycle completion popup	
<code>Enter</code>	Accept completion	
<code>,sn</code>	Edit snippets for filetype	

GIT (FUGITIVE)		
<code>,gs</code>	Git status	
<code>,gb</code>	Git blame	
<code>,gd</code>	Git diff split	
<code>,gS</code>	Open in GitHub	
<code>,gl</code>	Git log	
<code>:Gwrite</code> <code>:Gread</code>	Stage / checkout file	

INSIDE :GIT STATUS BUFFER		
<code>s</code> <code>u</code> <code>-</code>	Stage / unstage / toggle (git only)	
<code>=</code>	Toggle inline diff	
<code>cc</code> <code>ca</code>	Commit / amend	
<code>X</code>	Discard changes	
<code>dv</code>	Diff in vertical split	
<code>gq</code>	Close status buffer	

INSIDE :GDIFFSPLIT		
<code>do</code> <code>dp</code>	Diff obtain / diff put	
<code>]c</code> <code>[c</code>	Next / prev change	

GITGUTTER		
<code>:GitGutterNextHunk</code>	Next changed hunk	
<code>:GitGutterPrevHunk</code>	Prev changed hunk	
<code>:GitGutterStageHunk</code>	Stage hunk under cursor	
<code>:GitGutterUndoHunk</code>	Revert hunk under cursor	
<code>:GitGutterPreviewHunk</code>	Preview hunk diff	

L ^A T _E X (VIMTEX)		
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COMPILE & VIEW		
<code>-ll</code>	Compile (toggle continuous)	
<code>-lk</code>	Stop compilation	
<code>-lv</code>	View PDF in Skim	
<code>-lr</code>	Reverse search (PDF → source)	
<code>-le</code>	Errors to quickfix	
<code>-lc</code> <code>-lC</code>	Clean aux / clean all	
<code>-lt</code> <code>-lT</code>	Open / toggle TOC	
<code>-lo</code> <code>-li</code>	Compiler output / info	

NAVIGATION		
<code>[[</code> <code>]]</code>	Prev / next section	
<code>[m</code> <code>]m</code>	Prev / next environment	
<code>[n</code> <code>]n</code>	Prev / next math zone	
<code>[*</code> <code>]*</code>	Prev / next comment	

TEXT OBJECTS		
<code>ie</code> <code>ae</code>	Inside / around environment	
<code>ic</code> <code>ac</code>	Inside / around command	
<code>i\$</code> <code>a\$</code>	Inside / around math	
<code>id</code> <code>ad</code>	Inside / around delimiter	
<code>iP</code> <code>aP</code>	Inside / around section	
<code>im</code> <code>am</code>	Inside / around item	

CHANGE / DELETE SURROUNDING		
<code>cse</code> <code>dse</code>	Environment	
<code>csc</code> <code>dsc</code>	Command	
<code>cs\$</code> <code>ds\$</code>	Math	
<code>csd</code> <code>dsd</code>	Delimiter	

TOGGLE		
<code>tse</code>	Starred environment	
<code>tsc</code>	Starred command	
<code>ts\$</code>	Inline ↔ display math	
<code>tsf</code>	Inline ↔ display frac	
<code>tsd</code>	Delimiter modifier	

INSERT MODE		
<code>]]</code>	Close current environment	

FORTRAN		
<code>,:</code>	Align :: in declaration	
<code>,:</code>	Align all :: in file	

WINDOWS		
<code>Ctrl+Arrow</code>	Window in direction	
<code>,hs</code> <code>,vs</code>	Horizontal / vertical split	
<code>,sc</code>	Close window	
<code>,w=</code> <code>,wx</code>	Equalize / swap windows	
<code>Ctrl+W</code> + <code> </code> -	Taller / shorter	
<code>Ctrl+W</code> > <	Wider / narrower	
<code>Ctrl+W</code> H/J/K/L	Move window to edge	
<code>Ctrl+W</code> T	Window to new tab	
<code>Ctrl+W</code> o	Close all other windows	
<code>Ctrl+W</code> _	Maximize height / width	

BUFFERS & FILES		
<code>,z</code> <code>,x</code>	Prev / next buffer	
<code>Ctrl+Shift+Left</code> <code>Right</code>	Prev / next buffer	
<code>,.</code>	Toggle alternate file	
<code>,e</code> <code>,v</code>	Edit / view in current dir	
<code>,YY</code> <code>,YL</code>	Copy filepath / filepath:line	
<code>%%</code>	Current dir (cmd mode)	

PANELS & TOGGLES		
<code>\</code> <code>,nn</code>	Toggle NERDTree	
<code> </code> <code>,nf</code>	Find file in NERDTree	
<code>F8</code>	Toggle Tagbar	
<code>,ut</code>	Toggle Undotree	
<code>,t</code>	Terminal (bottom split)	
<code>Esc</code> <code>Esc</code>	Close terminal	

VINEGAR / NETRW		
<code>-</code> (<code>netrw</code>)	Open parent directory	
<code>.</code> (<code>netrw</code>)	File path at cmd line	
<code>y.</code> (<code>netrw</code>)	Yank absolute path	
<code>gh</code> (<code>netrw</code>)	Toggle hidden files	

INSIDE NERDTree		
<code>o</code> <code>Enter</code>	Open file / toggle dir	
<code>t</code> <code>s</code> <code>i</code>	Tab / vsplit / hsplit	
<code>p</code> <code>P</code>	Parent / root	
<code>r</code> <code>R</code>	Refresh dir / root	
<code>m</code>	Menu (create/rename/del)	
<code>I</code> <code>?</code>	Hidden files / help	
<code>C</code>	Change tree root to dir	
<code>u</code>	Root up one directory	

INSIDE TAGBAR		
<code>Enter</code> <code>p</code>	Jump / preview	
<code>o</code> <code>s</code>	Fold / sort	
<code>x</code> <code>q</code>	Zoom / close	

EUNUCH COMMANDS		
<code>:Rename</code> <code>:Move</code>	Rename / move file + buffer	
<code>:Delete</code> <code>:Copy</code>	Delete / copy file + buffer	
<code>:Mkdir</code> <code>:Chmod</code>	Create dir / change perms	
<code>:SudoWrite</code> <code>:Wall</code>	Write with sudo / write all	

QUICKFIX & LOCATION LIST		
<code>:copen</code> <code>:cclose</code>	Open / close quickfix	
<code>:cnext</code> <code>:cprev</code>	Next / prev quickfix entry	
<code>:lopen</code> <code>:lclose</code>	Open / close location list	
<code>:lnext</code> <code>:lprev</code>	Next / prev location entry	

USEFUL EX COMMANDS		
<code>:e!</code>	Reload file (discard changes)	
<code>:sort</code> <code>:sort u</code>	Sort lines / sort + unique	
<code>:retab</code>	Convert tabs ↔ spaces	
<code>:earlier 5m</code>	Undo to 5 minutes ago	
<code>:bufdo</code> <code>:cdo</code>	Cmd across buffers / quickfix	
<code>:LspInstallServer</code>	Install LSP for filetype	
<code>:LspStatus</code>	Running LSP servers	
<code>:LspStopServer</code>	Stop current server	

CONFIG & APPEARANCE		
<code>,vi</code> <code>,vl</code>	Edit vimrc / vimrc.local	
<code>,ss</code> <code>,sf</code>	Spellcheck / quick fix	
<code>,h</code> <code>,k</code>	Cursorline / syntax info	
<code>F1</code>	Show colorscheme	
<code>F2</code> <code>F3</code>	Prev / next colorscheme	
<code>F4</code> <code>F6</code>	Light-dark / paste mode	
<code>:W</code> <code>:Q</code> <code>:WQ</code> <code>:Wq</code>	Typo-safe save / quit	

FORMAT-ON-SAVE		
<code>Python</code> <code>.py</code>	ruff format (auto)	
<code>Fortran</code> <code>.f90</code> <code>.F90</code>	fprettify (auto)	
<code>C/C++</code> <code>,ff</code>	clang-format (manual)	
<code>ᲙᲗᲘ</code> <code>-ll</code>	latexmk (continuous)	